Jiaxuan Zou (邹嘉轩)

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Research Interests

Turning empirical training phenomena into first-principles mechanisms: mechanistic interpretability of LLMs, training dynamics of finite-width networks, optimizer design for LLM pre-training, scaling laws and their failure modes, deep learning theory and optimization.

Education

Xi'an Jiaotong University — Xi'an, China
B.S. in Mathematics and Statistics, School of Mathematics and Statistics — *Sep. 2024 – Present*

Research Experience

ByteDance, Seed Pre-training Team — Beijing, China
Research Intern — *Jul. 2026 – Oct. 2026*

Gaoling School of Artificial Intelligence, Renmin University of China — Beijing, China
Research Intern, advised by Prof. Yong Liu — *Jun. 2025 – Jun. 2026*

- Mechanistic interpretability, deep learning theory, optimizer design, and scaling laws.
- First-authored work on linear attention, neural scaling laws, and latent chain-of-thought.

ScalingOpt — Remote
Co-maintainer — *Apr. 2026 – Present*

- Benchmark and discussion platform for optimizer design in LLM training; maintain the optimizer library, including the Nora entry.

Tsinghua-affiliated AI Startup — Beijing, China
AI Technical Consultant

- Advise on "AI + K-12 Education" products developed with the Beijing Dongcheng District Education Commission.

Publications and Preprints

[1] **Jiaxuan Zou**, Ruifeng Ren, Yong Liu. **Kaczmarz Linear Attention.** Under review, 2026. arXiv:2605.08587.

[2] Jinghui Yuan, **Jiaxuan Zou**, Shuo Wang, Yong Liu, Feiping Nie. **Nora: Normalized Orthogonal Row Alignment for Scalable Matrix Optimizer.** Under review, 2026. arXiv:2605.03769.

[3] **Jiaxuan Zou**, Zixuan Gong, Ye Su, Huayi Tang, Yong Liu. **Effective Frontiers: A Unification of Neural Scaling Laws.** Under review, 2026. arXiv:2602.02593.

[4] **Jiaxuan Zou**, Yaozhong Xiong, Yong Liu. **Capabilities and Fundamental Limits of Latent Chain-of-Thought.** EMNLP, 2026. arXiv:2602.01148.

[5] Jialong Sun, Zeming Wei, **Jiaxuan Zou**, Jiacheng Gong, Guanheng Wang, Chengyang Dong, Jialong Li, Bo Liu. **Statistical MIA: Rethinking Membership Inference Attack for Reliable Unlearning Auditing.** Under review, 2026. arXiv:2602.01150.

[6] Jialong Sun, Xinpeng Ling, **Jiaxuan Zou**, Jiawen Kang, Kejia Zhang. **FreIE: Low-Frequency Spectral Bias in Neural Networks for Time-Series Tasks.** IEEE ICDM, 2025. arXiv:2510.25800.

[7] Tian Zhang, Jiaju He, Naihao Liu, **Jiaxuan Zou**, Yongxiang Jiang. **Mamba-Driven and Feature-Fused U-Net for Automatic Seismic Horizon Interpretation.** IEEE Transactions on Geoscience and Remote Sensing, 2026. doi:10.1109/TGRS.2026.3660880.

[8] **Jiaxuan Zou**, Naihao Liu, Tian Zhang, Jiaju He, Tao Li, Jinghuai Gao. **Seismic Horizon Picking Using Channel-Independent Multi-Scale UNet.** Under review, 2024.

Service and Outreach

Xi'an Jiaotong University Deep Learning Seminar — Xi'an, China
Founder and Organizer — *Sep. 2024 – Present*

- Founded an undergraduate deep learning seminar that grew to over one thousand participants nationwide and led to several research collaborations.

Skills and Languages

Mathematical tools	Optimization, statistics, dynamical systems, mathematical modeling
Applied AI	LLM pre-training, matrix optimizers, linear attention
Languages	Chinese (native), English (working proficiency)